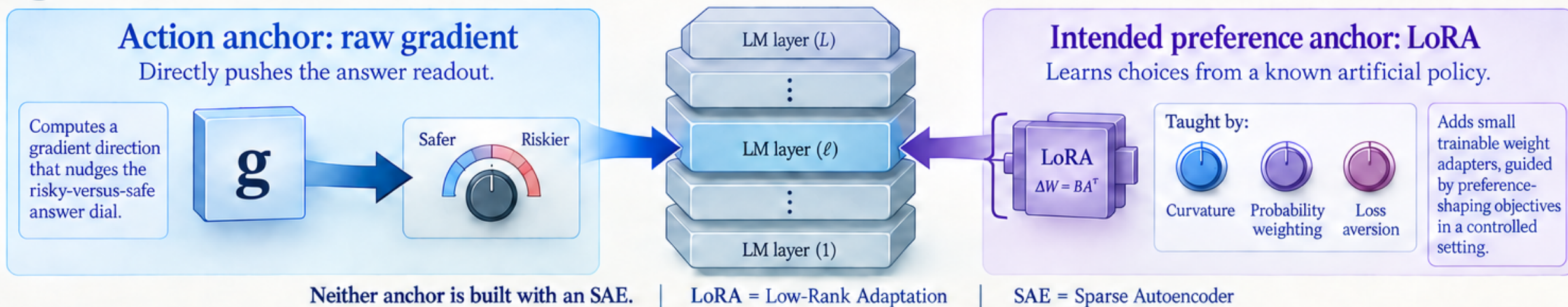


# Does steering change risk preferences?

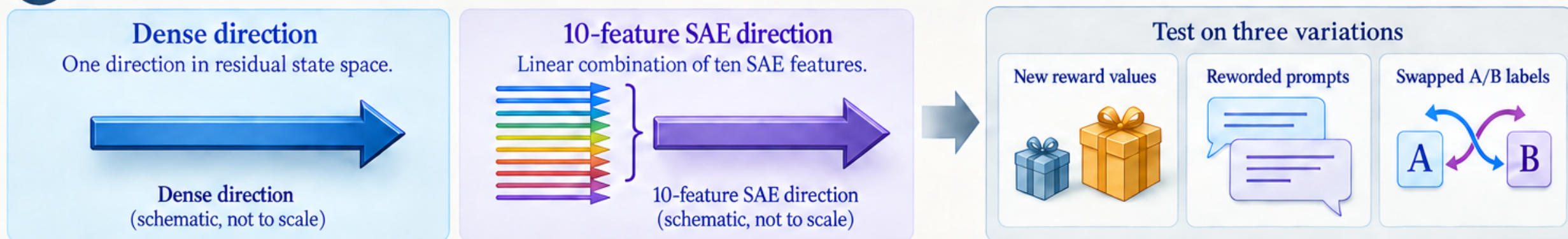
Or does it mainly change which answer the model selects?

## 1 Two independent reference interventions



## 2 Compare two ways to steer the same model

Same layer, positions and steering norm.



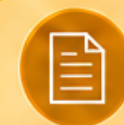
## 3 What the pilot supports



**LoRA:** recovery failed after rewording.



**Dense and SAE:** no complete preference-check passes.



**Current recipe not validated.**  
Broader SAE question remains open.

# What is Cumulative Prospect Theory?

CPT is a mathematical model of choices under risk.

## Option A

40 tokens for sure



## Option B

50% chance of 100 tokens  
50% chance of 0 tokens



The same gamble can appeal differently depending on how outcomes and probabilities are valued.

### 1 Outcome curvature



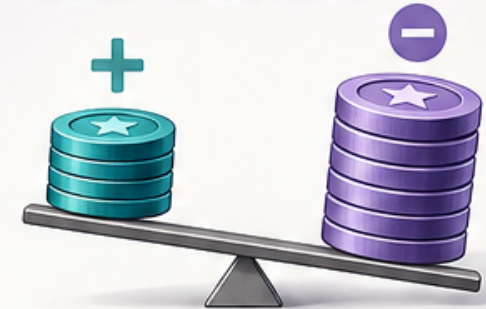
How subjective value changes as gains or losses grow.

### 2 Probability weighting



How decision weights differ from stated probabilities.

### 3 Loss aversion



How strongly losses count relative to gains.



Choose known CPT parameters



Generate artificial choices



Train a LoRA adapter  
LoRA = Low-Rank Adaptation



Test on new prompts



**A trained adapter is only an intended preference reference.**

Parameter recovery must hold on unseen rewards, reworded prompts and swapped answer labels.

# Changing choices is not yet a validated preference change

Question: can Sparse Autoencoder (SAE) steering produce a stable risk-choice policy across rewards, wording and answer labels?

**ACTION ANCHOR**

**Raw logit gradient**

A residual-space direction built from sensitivity of risky-versus-safe answer scores, averaged over discovery prompts. No SAE is used.

**Observed:** passed the action-control check at tested valid doses.

**INTENDED PREFERENCE ANCHOR**

**Low-Rank Adaptation**

LoRA is a trained weight-space adapter. We reused one trained to imitate a known Cumulative Prospect Theory (CPT) policy.

**Observed:** passed original-wording checks; failed rewording.

**METHODS EVALUATED**

**Dense and SAE steering**

A free residual direction versus a combination of ten SAE decoder features. Both aim at the same target policy, with matched edit size and positions.

**Observed:** neither method passed.

HELD-OUT RESULTS FOR THE COMBINED CPT TARGET

| Error relative to the intended policy | Original wording | New wording |
|---------------------------------------|------------------|-------------|
| LoRA reference                        | 0.042            | 0.115       |
| Dense steering                        | 0.129-0.151      | 0.224-0.225 |
| Ten-feature SAE steering              | 0.202-0.208      | 0.220-0.224 |

Root Mean Squared Error (RMSE) of choice probabilities; lower is better. Declared accuracy limit: 0.060. Steering ranges show two training seeds, not confidence intervals.

**Decision: defer the full feature sweep.** The tested recipe failed. A robust positive anchor and a passing dense control are still missing, so this does not establish a general limitation of the SAE dictionary.

# CPT gives us a known choice policy to aim for

CPT = Cumulative Prospect Theory. Here it is a mathematical behavioral benchmark, not a claim that a language model has human feelings.

**Illustrative question:** choose a guaranteed gain of 40 tokens, or a 50% chance of gaining 100 tokens and a 50% chance of gaining nothing. One answer cannot identify a preference; we vary outcomes and probabilities to measure a pattern.

## 1 / CURVATURE: ALPHA

### Sensitivity to amounts

How does subjective value change as gains or losses grow? Our combined target uses **alpha = 0.8**, giving diminishing sensitivity in magnitude. The same exponent is used for gains and losses.

## 2 / WEIGHTING: GAMMA

### Treatment of probabilities

Outcome probabilities are converted into decision weights using cumulative probabilities. Our combined target uses **gamma = 0.72**; gamma = 1 would leave probabilities undistorted.

## 3 / LOSS AVERSION: LAMBDA

### Relative weight of losses

How strongly does a loss count relative to an equal-sized gain? Our combined target uses **lambda = 2**: the value magnitude of a loss is twice that of the corresponding gain.

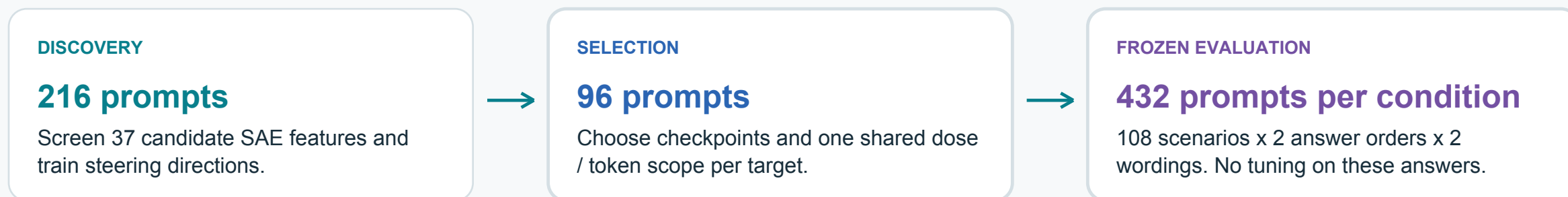
**Two additional fitted quantities:** choice sensitivity (inverse temperature, beta) controls how sharply value differences affect answer probabilities; an A-label bias captures a tendency to select the option displayed as A. Our CPT fit therefore has **five parameters**, not only three.

**Two steering targets:** combined = (alpha 0.8, gamma 0.72, lambda 2); neutral = (1, 1, 1). Both use beta = 3 and zero A-label bias. The neutral target is linear and probability-unweighted, but choices remain probabilistic.

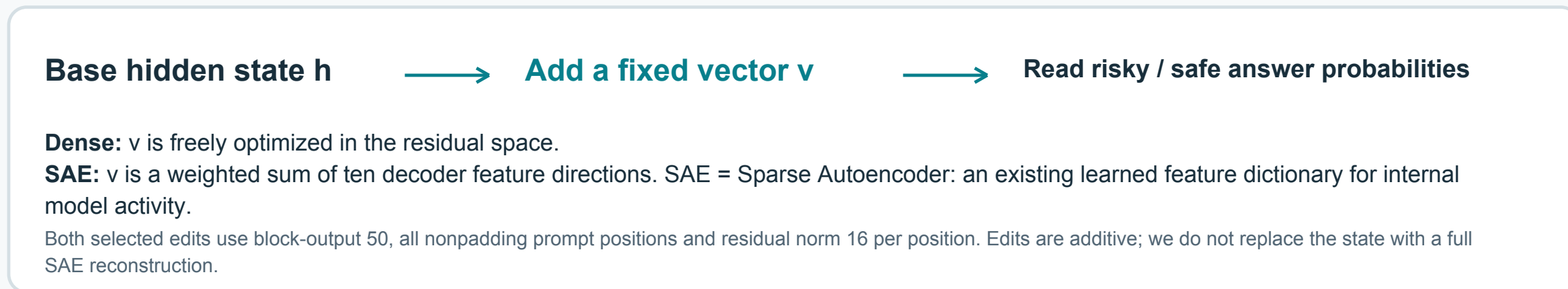
Implementation: outcomes are changes from a zero reference, scaled by 100 tokens. Gain/loss curvature and probability weighting are shared in this restricted CPT family. It does not cover every preference model. The example above is illustrative, not a measured test row.

# Learn on development prompts; judge on untouched prompts

LLM = Large Language Model. We used the same Llama-3.3-70B-Instruct checkpoint in bfloat16 (bf16), a 16-bit numerical format.



## WHAT WAS MATCHED BETWEEN DENSE AND SAE STEERING



**What matching does and does not mean:** the two methods receive the same-sized residual displacement at the same locations. That does not guarantee equal behavioral strength or optimization difficulty. The weight-space LoRA cannot be norm-matched this way.

**Other controls:** the unmodified model, reused LoRA, raw positive/negative readout gradients, sparse SAE readout approximation, random directions, and 48 elementary arithmetic/probability/dominance items. There were 28 final conditions in total.

Seeds are independent optimization initializations, not independent base models. Feature screening ranked full-grid choice loss, not direct switching-point / curvature effects. No new SAE or LoRA was trained in this latest round.

# Why we used the phrase "bias-like"

This is a held-out predictive comparison between two explanations of the observed SAE-steered choices. It is not a semantic label for an SAE feature.

EXPLANATION A / 5 FITTED PARAMETERS

**Restricted CPT model**

Predict choices from rewards and probabilities using curvature, weighting, loss aversion, choice sensitivity and A-label bias.

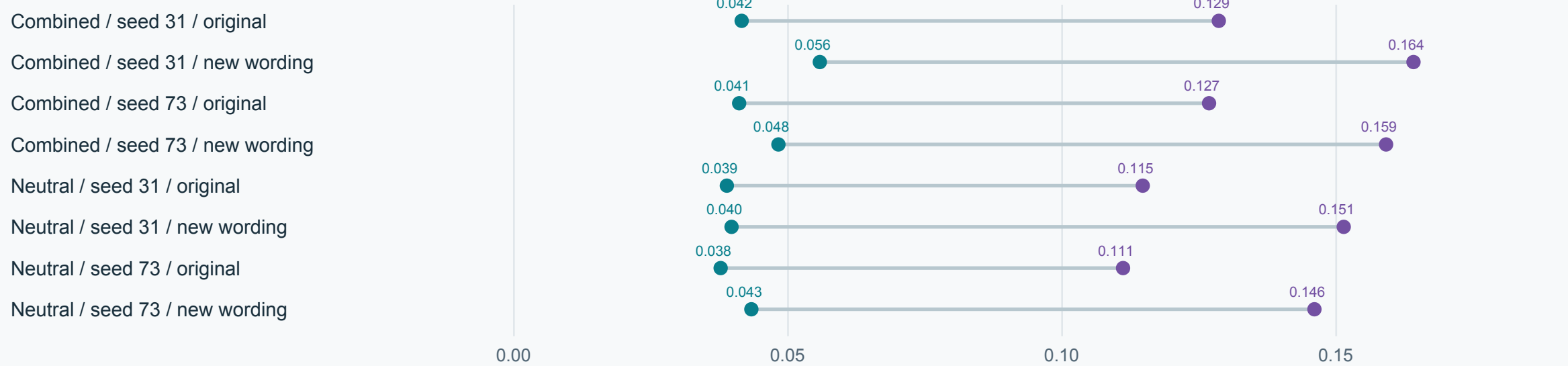
EXPLANATION B / 7 FITTED PARAMETERS

**Answer-bias / confidence model**

Transform the base model's risky/safe log-odds with a separate shift and positive scale for each of three frames, plus an A-label bias.

**Both explanations were fitted on selection responses and then frozen.** The chart compares their error when predicting the actual SAE-steered probabilities on unseen scenarios and wording.

HELD-OUT EXPLANATORY RMSE / LOWER IS BETTER



**Interpret carefully:** the bias model has more parameters and access to base-model responses. Better held-out prediction supports this descriptive explanation; it does not prove an internal mechanism or exclude every preference model.

# The current recipe failed; the general SAE question stays open

Keep target recovery, behavioral model comparison, and feature meaning as three separate claims.

QUESTION 1 / DID THE INTERVENTIONS RECOVER THE PLANTED POLICY?

| Target   | Intervention     | Original wording | New wording   |
|----------|------------------|------------------|---------------|
| combined | LoRA             | 0.0416           | 0.1147        |
| combined | Dense vector     | 0.129-0.151      | 0.224-0.225   |
| combined | Ten SAE features | 0.202-0.208      | 0.220-0.224   |
| neutral  | Dense vector     | 0.1319-0.1323    | 0.241-0.248   |
| neutral  | Ten SAE features | 0.2035-0.2043    | 0.2532-0.2533 |

RMSE against the planted teacher; target <= 0.060. Ranges span two seeds, not uncertainty intervals. These are different errors from page 4, whose target was the actual steered behavior.

ACTION REFERENCE

**Raw gradient: pass**

The valid-dose raw gradient met the operational action-control criterion. That validates one useful negative control.

PREFERENCE REFERENCE

**LoRA: partial success**

Passed the latest provisional criteria in original wording. Rewording failed recovery, prediction and answer-order checks.

STEERING OUTCOME

**Dense 0/8; SAE 0/8**

Eight combinations = 2 targets x 2 seeds x 2 wordings. None passed the complete rule; this is not eight separate tests.

**Beyond exact target recovery:** a separate preference-like check removed the planted-parameter and teacher-accuracy requirements. Both methods still passed 0/8. Yet failure of a finite candidate set and optimizer budget is not evidence that every SAE intervention must fail.

The earlier stricter repair rule used teacher RMSE <= 0.040; the reused LoRA misses it on this new grid in both wordings. "Pass" above refers only to the latest provisional rule. These thresholds do not replace the collaborator's full go/pivot protocol.

# What to align with coauthors before another GPU run

The freely trained dense vector is a new diagnostic control. The original probe is also a dense residual direction, learned in a different way.

|                   | Current dense control                       | Original Llama probe code                |
|-------------------|---------------------------------------------|------------------------------------------|
| How learned       | Optimize CPT choice-probability loss        | Train a linear classifier on activations |
| Where applied     | Block-output 50; selected all prompt tokens | Layer 48; final 20% with a position ramp |
| Comparison status | Evaluated in this pilot                     | Not rerun under these matched conditions |

We cannot infer how the original probe would perform from this dense control's failure. A fair head-to-head needs aligned prompts, layer, token scope, dose, training objective and held-out criteria.

INTERPRETABILITY STATUS

Existing dictionary; our analyses

We used Goodfire's released layer-50 SAE. Descriptions are cached Neuronpedia labels, not independently validated meanings. Activations and steering effects are our measurements.

SPARSE READOUT DIAGNOSTIC

Large residual is scope-limited

The ten-feature readout approximation left relative residual norm 0.961. This diagnoses that sparse fit; it does not show that the full dictionary cannot express the gradient.

**Next gate:** validate a LoRA across new phrasings, label orders and reward grids. If it passes, repeat the small matched dense / SAE comparison. If dense succeeds and SAE fails, the sparse recipe is implicated more specifically. If both fail, change the intervention format before a full sweep.

**Definitions and sources:** CPT: Tversky & Kahneman (1992), [doi:10.1007/BF00122574](#). LoRA: Hu et al. (2021), [arXiv:2106.09685](#). SAE: [Goodfire layer-50 model card](#). Result source: decision-refined-20260918T235147Z, decision\_analysis.json. Local source index, full methods and figure data accompany this brief.

Prepared with Codex using saved results checked against the analysis code. This brief adds no experimental data; its figures are drawn from the saved analysis. The previous GPU session was stopped and its artifacts verified.